

CLL Agent — STT & TTS Challenges

by AI Team · BitLogix

Speech-to-Text (STT) Challenges

1. Audio Quality: 8 kHz Telephony vs 16 kHz ASR

Factor	Detail
Recorded sample rate	8 kHz (telephony standard)
Model requirement	16 kHz (Whisper, Qwen3-ASR, all major ASR models)
Problem	Upsampling from 8 → 16 kHz adds no information — high-frequency band (4–8 kHz) containing Urdu fricatives (غ, خ, ف, ز, س, ش) is permanently lost at the codec stage
Impact	Higher WER on fricative-heavy Urdu medical vocabulary; muffled/distorted signal degrades confidence across the board

Mitigation options: Request 16 kHz recording from PBX before codec encoding; use telephony-trained acoustic model variant; apply spectral enhancement pre-processing.

2. Dual-Speaker Audio: Agent + Caller on One Channel

Speaker	Characteristics	Challenge
Caller (patient)	Normal volume, variable pace, may have accent	Natural variation — handled reasonably by model
Agent (helpdesk staff)	Low volume (distant mic), very fast speech (experienced professional cadence), domain jargon	Under-represented in training data for fast low-volume speech
Combined	Single mixed audio channel	No speaker separation; ASR sees both speakers as one signal

Impact: Agent utterances get transcribed with higher error rate; fast speech causes word merging; low-volume agent audio may be partially drowned by caller.

Mitigation options: Separate recording channels (two-channel PBX recording); voice activity detection + volume normalisation; diarisation before ASR.

3. Language-Hint Dilemma: Hindi vs English Prompt

The ASR model (Qwen3-ASR) requires a language hint in its prompt. Neither available option is correct for Urdu:

Language Hint	Behaviour	Problem
language Hindi	Outputs Hindi Devanagari script; preserves Urdu word content	Drops nuktas (◌ِ) — क→क, ज→ज, फ→फ — so قابل→kaabil not qaabil, زندگی→jindagi not zindagi
language English	Preserves nuktas; phonetically more accurate Devanagari	Transcribes code-switched or ambiguous segments in Latin/English — entire Urdu phrases can come out in English

Root cause: No **language Urdu** option — Urdu is not in the model's supported language list. Urdu uses Perso-Arabic script; the model transcribes it as Hindi/Devanagari. Nuktas distinguish Perso-Arabic phonemes but Hindi training data omits them.

Mitigation options: Fine-tune model on Urdu audio with nukta-annotated Devanagari targets; post-process with rule-based nukta restoration (ph→f closed set + lexicon); submit Urdu support request upstream.

4. Code-Switching and Medical Vocabulary

- Callers mix Urdu, English medical terms (blood test, CBC, ultrasound, creatinine) and sometimes Punjabi
- Medical terminology (lab test names, drug names, anatomical terms) is out-of-distribution for general ASR models
- Names: doctor names, test names, branch names (Johar Town, Gulberg, Liberty) are frequently mis-transcribed

5. STT Model Training Challenges

Lack of Urdu-specific training data: - Most open-source ASR models (Whisper, Qwen3-ASR) are trained predominantly on English and high-resource languages; Urdu is severely under-represented - No large-scale, clean, transcribed Urdu telephony corpus exists publicly — CommonVoice Urdu is small (~100 hours) and read-speech only, not conversational - Collecting and annotating domain-specific Urdu helpdesk audio (medical terms, lab tests, proper nouns) requires significant manual effort and native-speaker expertise

Fine-tuning complexity: - Fine-tuning Whisper or Qwen3-ASR on Urdu requires GPU infrastructure, careful learning rate scheduling, and risk of catastrophic forgetting on other languages - Urdu ground-truth transcripts must be in Nastaliq script with consistent orthographic conventions — inconsistency in training labels directly causes WER regression - Evaluation requires native Urdu annotators; standard WER metrics do not account for script normalisation (e.g., alef variants: اَ، آ، اِ)

Domain mismatch: - General ASR models are trained on news, podcasts, and read speech — not live telephony conversations with interruptions, hesitations, and background noise - Medical vocabulary (haematology, biochemistry, radiology terms) is out-of-vocabulary in all general models; fine-tuning on in-domain data is essential but expensive - Accents across Pakistani cities (Lahore, Karachi, Peshawar, Faisalabad) are not uniformly represented — model generalises poorly to under-represented regional accents

6. Progress: Hindi → Roman Urdu

Since no native Urdu ASR model exists on-premises, the current approach bridges the gap through a custom post-processing layer built at BitLogix:

How it works:

1. **ASR transcribes in Devanagari** — Qwen3-ASR outputs Hindi/Devanagari text for Urdu speech (e.g., "زندگی" is transcribed as "ज़िंदगी")
2. **Phoneme mapping (Layer 1)** — Each Devanagari character is mapped to its Roman Urdu equivalent using a hand-crafted consonant and vowel map; nukta characters (़) are resolved to their Perso-Arabic phonemes (फ़→f, ज़→z, क़→q)
3. **Schwa deletion (Layer 2)** — Hindi inherits Sanskrit schwa (implicit /ə/ vowel) which is silent in many Urdu words; rule-based schwa deletion removes it to produce correct Roman spelling (e.g., "करम" → "karam" not "karama")
4. **Lexicon correction (Layer 3)** — A domain lexicon of ~985 word corrections and ~286 proper nouns fixes residual errors that rules cannot handle (medical terms, Urdu-specific phonology, names)

What it solves:

Problem	Solution
No native Urdu ASR	Devanagari output re-mapped to Roman Urdu — downstream NLP works on Roman script
Nukta omission (Hindi hint)	English language hint used on CPU — better nukta retention; nukta-aware phoneme map handles फ़/ज़/क़ correctly
Urdu-specific phonology (ph→f, j→z)	Closed rule sets handle systematic substitutions; open sets handled by lexicon
Medical and proper nouns	Proper nouns dictionary covers lab branch names, doctor names, test names
Script mismatch for NLP	Output is clean Roman Urdu — compatible with intent classifiers and LLM context

Current limitations: Lexicon must be maintained manually as new vocabulary is encountered; regional accent variation can produce Devanagari forms that miss the lexicon; nukta emission remains inconsistent across audio quality levels.

Text-to-Speech (TTS) Challenges

1. Telephony Downsampling: 16 kHz → 8 kHz Output

- TTS models generate audio at 16–24 kHz (full bandwidth, natural-sounding)
- PTCL SIP / G.711 delivers to caller at 8 kHz
- Downsampling collapses the high-frequency band: synthesised voice loses naturalness, sounds muffled over phone
- Urdu fricatives (ش، س، ز، خ) are especially degraded — clarity of dental and emphatic consonants reduced

2. Scarcity of High-Quality On-Premises Urdu Models

Tier	Examples	Gap
Cloud (excluded)	Google WaveNet Urdu, Azure Neural Urdu, Amazon Polly	Privacy / on-prem policy
Research models	IMS-Toucan (multilingual), MMS-TTS	Urdu quality inconsistent; not production-ready
Open on-prem	Coqui XTTS (zero-shot), Piper TTS	Limited Urdu training data; accent/naturalness issues
Target	VITS or XTTS fine-tuned on Urdu	Requires custom dataset + GPU training

Fine-tuning a neural TTS requires 5–20 hours of clean, professionally recorded Urdu speech — not readily available.

3. Missing Diacritics (Aerab) Problem

- Urdu Nastaliq script is routinely written **without aerab** (short vowels: پیش، زیر، زبر) in all practical text sources
- Without aerab, the same written word has multiple valid pronunciations — TTS must infer the correct one from context
- Example: "کرم" = karam (grace) or kirm (worm) depending on context; TTS with no aerab will guess
- LLM-generated response text will always lack aerab — no practical way to enforce diacritic annotation at scale

Mitigation: Diacritic prediction model (grapheme-to-phoneme with context); constrained vocabulary for known domain terms.

4. Mixed-Language Pronunciation and Script Conflicts

- Response text from LLM contains Urdu (Nastaliq), English terms, numbers, and sometimes Devanagari
- TTS must correctly pronounce English words embedded in Urdu (e.g., "CBC test کے لیے Johar Town آئیں")
- Naive single-model TTS applies Urdu phonology to English words → heavy foreign accent on medical terms
- Solution requires per-token language ID + multilingual TTS or a separate English synthesis pass with prosodic blending

5. Number, Price, and Date Normalisation

- LLM outputs may include: **Rs. 1,450**, **12 May 2026**, **10:30 AM**, **CBC (Complete Blood Count)**
- TTS requires normalised text-to-speech form: "ایک ہزار چار سو پچاس روپے", "بارہ مئی دو ہزار چھبیس"
- No robust Urdu text normalisation library exists; must be built or adapted from multilingual normalisation tools
- Edge cases: ranges ("500–1500"), abbreviations ("Dr.", "Dept."), mixed units

6. Naturalness, Prosody, and Real-Time Latency

- Conversational helpdesk requires TTS response within ~1 second of intent completion
- Neural TTS batch-processes full sentences — streaming requires chunked synthesis with sentence boundary detection
- Helpdesk callers are patients — empathetic, clear, unhurried delivery matters for trust
- Current on-prem Urdu models produce flat monotone speech; sentence-final intonation for questions vs statements is often incorrect

Model Type	RTF (approx.)	GPU Requirement	Notes
HMM/Concatenative	< 0.1×	CPU only	Robotic; acceptable latency
VITS (fine-tuned)	~0.3–0.5×	Mid-range GPU	Best quality/speed tradeoff
XTTS v2 (zero-shot)	~1–2×	8 GB VRAM	High quality; slow without batching
Large diffusion TTS	> 2×	High-end GPU	Not viable for real-time

7. No Custom Brand Voice

- Chughtai Lab may require a consistent, recognisable helpdesk voice identity
- Custom voice cloning (XTTS speaker conditioning, VALL-E style) requires 5–30 minutes of target speaker recordings
- Voice consistency across sessions, call types, and languages must be maintained
- Requires a dedicated voice recording session with a professional Urdu speaker

Prepared for Chughtai Lab AI Helpdesk · Chughtai Lab Voice Pipeline Technical Review